

WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Session 14: Unsupervised Learning & GMM

K-Means, Mixture Models, and Expectation-Maximization (EM)

Read alongside the lecture slides · Every slide example worked out in full

1 Introduction to Unsupervised Learning

Discovering hidden patterns and structures in unlabeled data without target ground truth.

In supervised learning, every training sample comes with a known target label y . In **unsupervised learning**, we receive only feature vectors $x_1, x_2, \dots, x_N$ without ground truth labels. The goal is to discover latent structure, underlying clusters, or generative probability distributions.

The intuition

Unsupervised learning is like sorting a large pile of mixed coins into groups based on size and weight without knowing coin names.

2 K-Means Clustering

Partitioning N samples into K disjoint clusters by minimizing within-cluster squared distances.

K-Means is a hard-clustering algorithm that partitions data into K distinct clusters centered at means $\mu_1, \dots, \mu_K$.

2.1 Objective Function and Algorithm

K-Means minimizes the distortion measure J :

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|x_n - \mu_k\|^2 \quad (1)$$

where $r_{nk} = 1$ if sample x_n is assigned to cluster k , and $r_{nk} = 0$ otherwise.

1. **E-Step (Assignment):** Assign sample x_n to the nearest centroid μ_k :

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|x_n - \mu_j\|^2 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

2. **M-Step (Update):** Recompute centroids by averaging assigned points:

$$\mu_k = \frac{\sum_{n=1}^N r_{nk} x_n}{\sum_{n=1}^N r_{nk}} \quad (3)$$

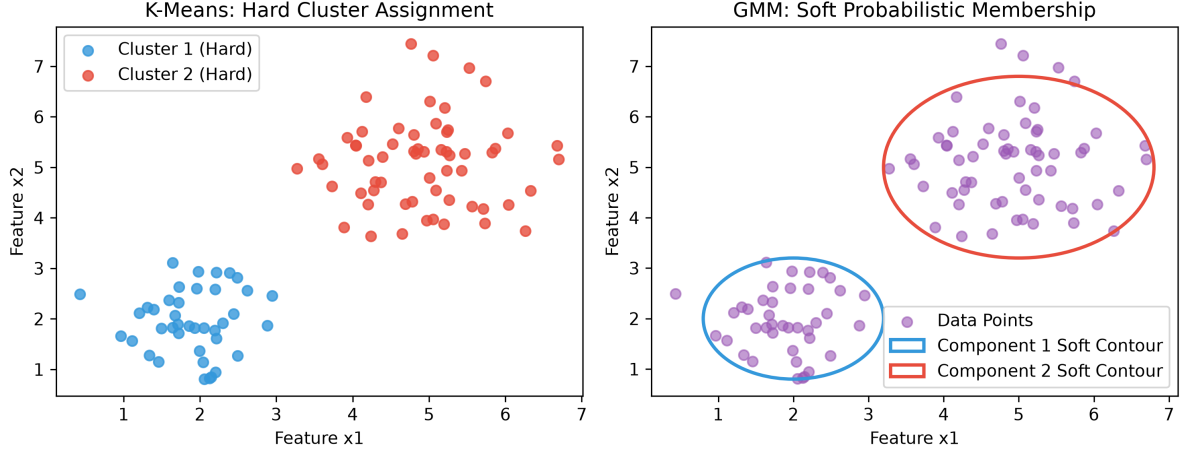


Figure 1: Comparison of hard K-Means boundaries versus soft Gaussian Mixture Model probabilistic boundaries.

3 Gaussian Mixture Models (GMM)

Soft probabilistic clustering representing data as a weighted mixture of K Gaussian distributions.

While K-Means makes hard binary assignments, a **Gaussian Mixture Model (GMM)** calculates the soft probability (responsibility γ_{nk}) that point x_n belongs to Gaussian component k .

3.1 GMM Formulation

The overall probability density function is a linear combination of K Gaussians:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k) \quad (4)$$

where π_k are mixture weights satisfying $\sum_{k=1}^K \pi_k = 1$ and $\pi_k \geq 0$.

4 The Expectation-Maximization (EM) Algorithm

Iterative optimization framework for finding maximum likelihood parameter estimates in latent variable models.

4.1 EM Algorithm Steps

Step 1. Initialize: Set initial parameters μ_k, Σ_k, π_k .

Step 2. Expectation Step (E-Step): Calculate responsibilities γ_{nk} :

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_n | \mu_j, \Sigma_j)} \quad (5)$$

Step 3. Maximization Step (M-Step): Re-estimate parameters using γ_{nk} :

$$N_k = \sum_{n=1}^N \gamma_{nk}, \quad \mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} x_n, \quad \pi_k^{\text{new}} = \frac{N_k}{N} \quad (6)$$

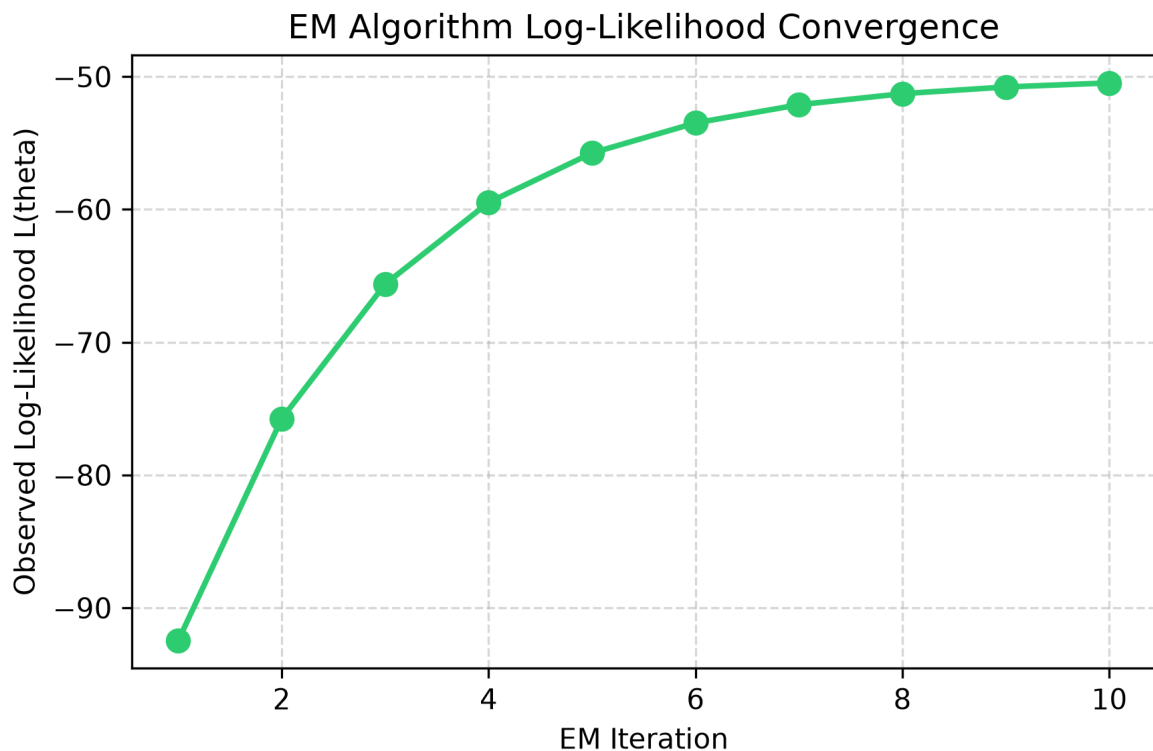


Figure 2: The EM algorithm monotonically increases the log-likelihood $L(\theta)$ until convergence.

Worked example: GMM Numerical Calculation

Let us execute the GMM numerical slide example step-by-step:

Consider 1D data points $X = [1, 2, 5, 6]$. We fit $K = 2$ Gaussians.

Initial parameters: $\mu_1 = 1.5$, $\mu_2 = 5.5$, variances $\sigma_1^2 = 1$, $\sigma_2^2 = 1$, weights $\pi_1 = 0.5$, $\pi_2 = 0.5$.

Step 1. E-Step for $x_1 = 1$: Compute $\mathcal{N}(1 | 1.5, 1) = \frac{1}{\sqrt{2\pi}} e^{-0.5(1-1.5)^2} = 0.352$.

Compute $\mathcal{N}(1 | 5.5, 1) = \frac{1}{\sqrt{2\pi}} e^{-0.5(1-5.5)^2} = 0.000014$. Responsibility $\gamma_{1,1} = \frac{0.5 \times 0.352}{0.5 \times 0.352 + 0.5 \times 0.000014} \approx 0.9999$.

Step 2. M-Step Update for μ_1 : Update mean $\mu_1^{\text{new}} = \frac{\sum_n \gamma_{n1} x_n}{\sum_n \gamma_{n1}} \approx \frac{1(1) + 1(2) + 0(5) + 0(6)}{1 + 1 + 0 + 0} = 1.5$.

5 Summary & Self-Test

✓ Key takeaway

K-Means performs hard clustering by minimizing squared distance. GMM extends this to soft probabilistic clustering using the EM algorithm, updating responsibilities (E-step) and Gaussian parameters (M-step).

5.1 Self-Test Quiz

Q1: What is the main difference between K-Means and GMM?

Answer: K-Means makes hard assignments, whereas GMM computes soft probabilistic responsibilities.